

Vinay Jampana

Senior Software Engineer – AI & Full-Stack

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Summary

AI full-stack engineer with 5 years across Node.js/NestJS, Python, React/TypeScript and AWS. I design and ship production AI agents end to end, with the quality checks and monitoring that keep them reliable. AI engineering since 2024.

Skills

Backend: Node.js, Python, TypeScript, NestJS, FastAPI, PostgreSQL

Agentic AI: LLM agents, LangGraph, RAG, LiteLLM, OpenSearch, Langfuse

Frontend: React, Next.js

Cloud: AWS, Docker, GitHub Actions

Experience

Zotok.ai – Senior Software Engineer (AI & Full-Stack)

May 2023 – Present

Stack: Node.js, Python, NestJS, LangGraph, PostgreSQL, OpenSearch, React, AWS

AI engineering

- Designed an **agent orchestration** platform (LangGraph, NestJS): one black box became seven testable steps on a visual builder over 50 typed functions, so per-customer changes ship without code or a redeploy
- Designed the **multi-agent** layer of the WhatsApp sales agent (LangGraph): messages route between service and payment agents, and a **guardrail** classifier filters spam, auto-replies and off-topic chat before the model answers
- Built an **LLM evals** system for the agent: 64 scenarios graded line by line, run twice daily with Teams alerts; it repeatedly caught breakages from other teams' backend changes before production
- Built the agent's **RAG** pipeline (OpenSearch vector search, LiteLLM embeddings) with **multi-tenant** isolation, so every customer's answers come only from their own documents
- Built a matching engine that maps shorthand user text to catalog items over 15,242 production aliases: the correct first match rose **69–78%** in testing and **81–95%** on real user messages
- Built a **multimodal** extraction pipeline: photos of handwritten sheets become structured carts through OCR and entity matching, at **88%** line accuracy on a 378-line test set
- Applied **prompt engineering** per customer: workspace guidance is injected into extraction, matching and OCR prompts, so each customer's terminology works without code changes
- Built an **LLM gateway** integration (LiteLLM) with per-workspace keys and spend tags per feature, so model cost is tracked by customer and feature

Backend engineering

- Designed and built a **WhatsApp workflow** (agent, NestJS backend, React web app) that replaced a plant's offline pour and dispatch forms; over 3,800 pour cards and 1,900 dispatches captured since launch
- Designed **human-in-the-loop** escalation for the WhatsApp assistant: intents the bot should handle run at once while others wait for a human reply, with per-conversation ordering through message groups
- Built the typed service behind the agent (NestJS, PostgreSQL): over 50 functions on one shared contract, with product search **4x faster** (875 to 228 ms per line)

Frontend engineering

- Built dynamic WhatsApp template buttons: five button types resolve at send time into **deep links** to the right page for each buyer, from the template builder to the landing page
- Cut first page load from 15–30 seconds to **1.5 seconds** by removing unused mobile code, rebuilding the Webpack config and lazy loading; builds got 30–40% faster

Sumeru Software – Senior Developer

Aug 2022 – Jan 2023

- Built a role-based CRM from scratch (React, Redux Toolkit, Node.js) with permission-based screens and modular dashboards; owned it from kickoff to production and shipped the MVP in **4 months** with a team of 4

Aerchain – Software Development Engineer

Intern Jul 2019 – Jul 2020; SDE 1 Apr 2021 – Jan 2022

- Built a real-time auction module for a B2B procurement platform with an **optimistic UI** that reconciles with the server, and barcode-scan inventory management as an **offline-first** PWA

Education

B.E. (Hons) Mechanical Engineering, BITS Pilani

2016 – 2020